

EMBEDDED SYSTEMS ENGINEER III

Hardware Interaction | Firmware | Embedded Systems

Hands-on experience across mechanical design, electronics, firmware, UX and system integration, developing consumer electronics, VR and medical devices from concept and prototype through manufacture, deployment and support. Focused on integrating physical design, embedded systems and software to deliver robust, production-ready products beyond the first demo.

EXPERIENCE III

Valkyrie Industries || Electronics Engineer | London, Sep 2024-Jun 2026

Developed *eir bands*, a neuromuscular stimulation device for stroke rehabilitation, while supporting production, procurement, factory oversight, fulfilment and IEC 60601 testing with UL.

- Owned firmware and hardware development with a focus on robustness, reliability and commercial use.
- Developed a closed-loop stimulation system using an nRF-based IMU to trigger stimulation while relaying device status and orientation data to a BLE client.
- Built device-mobile app communication, app integrate, and firmware deployment/update systems.
- Designed bed-of-nails fixtures, development jigs and quality-control tools for in-house testing and mass manufacture, together with the required SOPs and test criteria.
- Managed BOMs, procurement, and communication with manufacturers.
- Oversaw production processes and build quality including factory visit in China.
- Supported hardware testing with UL toward IEC 60601 medical-device certification.
- Developed a repeatable workflow covering firmware updates, final testing, packaging, shipment preparation and logistics coordination to streamline fulfilment of 600 devices to customers.

TGo || Design Engineer | London, Jan 2019-Sep 2023

Worked across [consumer, automotive and medical sectors](#) using capacitive touch and pressure-sensing technologies, with a main focus on the [eteerX](#) family of finger-tracking VR controllers.

Product Development, Systems Integration, Manufacturing & Production

- Worked across the full product lifecycle from concept, physical design and ergonomics through firmware, software integration, manufacture, release and customer support, with readiness for scaling.
- Managed software and electronics development around hardware interaction and UX improvements.
- Developed device driver using the OpenVR SDK to integrate the controller with SteamVR systems.
- Reverse-engineered reference hardware, interfaces and haptic circuits to understand system behaviour and support compatibility and future development.
- Designed assembly and test jigs, manufacturing software and SOPs for production and quality control.
- Designed products for manufacturability, from concept through to mass production,

Firmware & Electronics Development, R&D, Leadership & Client Support

- Developed firmware algorithms mapping capacitive inputs to LED and haptic feedback, alongside continuous feature development, debugging and bug fixing.
- Improved haptic performance across TGo products and supported robustness for consumer release and post-release firmware updates.
- PCB design and electronics debugging, modification and rework for through-hole and SMD components.
- Led R&D across hardware, firmware, software, and testing while coordinating team members, resources and project delivery.
- Managed B2B product development including client communication & satisfaction.
- Technical lead for customer support, issue investigation and internal workflow improvements.

SKILLS III

Embedded Systems, Firmware & Electronics

- BLE/ nRF52/ ESP32/ ESP-NOW/ Git/ Version control/ UART, I²C and SPI/ Oscilloscope and logic analyser use/ Hardware debugging
- Embedded C, C++, and Arduino
- Firmware development, debugging and hardware integration
- KiCad and PCB design
- Through-hole and SMD hand soldering, prototyping, testing and rework

Mechanical & Digital Design

- SolidWorks/ Onshape/ Blender
- 3D printing/Workshop fabrication using plastics, metal and wood
- Jig and fixture design/ DFM/ DFA
- Adobe Illustrator, Photoshop, InDesign and Premiere Pro